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Student Number

2016

Mathematics

HSC Course Assessment Task 3

1st June 2016

General Instructions

- Reading time – 5 minutes
- Working time **60 minutes**
- Write using blue or black pen
Black pen is preferred
- Board approved calculators may be used
- In Questions 4 - 16 show relevant mathematical reasoning and/or calculations
- Use a separate booklet for each question
- A reference sheet is provided at the back of this paper
- Diagrams are NOT to scale (unless indicated)

Total Marks – 42

Section I - Page 2

3 marks

- Attempt Questions 1 - 9
- Allow about 5 minutes for this section

Section II - Pages 3 - 5

41 marks

- Attempt Questions 4 - 6
- Allow about 55 minutes for this section

Multiple Choice	/ 3
Q 4 (Logarithms and Exponentials)	/ 21
Q 5 (Probability)	/ 7
Q6 (Applications of Calculus to the Physical World)	/ 11
Total	/ 42

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 20% of the HSC Course Assessment

Section I

3 marks

Attempt Questions 1 - 3

Allow about 5 minutes for this section

Use the multiple-choice answer sheet for questions 1 -3 (Detach from paper)

1. Given that $\log_a x = 0.3$ and $\log_a y = 0.2$, then the value of $\log_a x^2 y^3$ is
- (A) $a^{0.36}$
(B) 1
(C) 1.2
(D) $a^{0.0036}$
2. If $y = e^x - e^{-x}$, then the gradient of the tangent at $x = 0$ is:
- (A) 0
(B) 2
(C) -2
(D) e^2
3. At Killara High School the probability that a student takes a Mathematics course is 0.9 and the probability that a student takes a Science course is 0.8. The probability that a student takes a Mathematics or a Science course is 0.95.

What is the probability that a student does both Mathematics and Science?

- (A) 0.1
(B) 0.2
(C) 0.75
(D) 1

Section II

39 marks

Attempt Questions 4 - 6

Allow about 55 minutes for this section

Answer each question in a separate booklet. Extra writing booklets are available.

In questions 4 – 6 your responses should include relevant mathematical reasoning and / or calculations.

Question 4 (21 marks) Use a SEPARATE writing booklet

(a) Differentiate with respect to x and fully simplify

i) $y = e^x \log_e x$ 2

ii) $y = \cos(\log_e x)$ 2

(b) Evaluate using exact values $\int_0^{e^2} \frac{4}{x} dx$ 3

(c) Solve for x 3
$$e^{2x} - 2e^x = 0$$

(d) Let $y = \log_e(x - 1)$

i) Find the x – intercept. 2

ii) Sketch the graph of $y = \log_e(x - 1)$. 2

iii) Find the exact value of the area bounded by $y = \log_e(x - 1)$, x – axis and the line $x = 5$. 4

iv) Find the exact volume of the solid generated when the area between $y = \log_e(x - 1)$, x – axis and y – axis is rotated about the y – axis. 3

Question 5 (7 marks) Use a SEPARATE writing booklet

- a)** Two standard six sided dice are tossed. Given that a 4 is showing on the upmost face of one of the dice , what is the probability that the sum of the upmost faces on both dice exceeds 8? **2**

- b)** In a game of chess between two players, X and Y, of about equal ability, the player with the white pieces, having the first move, has a probability of 0.5 of winning, and the probability that the player with the black pieces wins is 0.3.

The two players, X and Y, play each other twice in a chess competition, each player having the white pieces once. In the competition, a player who wins a game scores 1 point, a draw scores as $\frac{1}{2}$ a point to each player and a player who loses scores 0 points.

- i)** What is the probability that the game ends in a draw (i.e. no player wins) **1**
- ii)** What is the probability that, as a result of these two games X scores 2 points? **2**
- iii)** What is the probability that, as a result of these two games X scores $1\frac{1}{2}$ points? **2**

Question 6 (11 marks) Use a SEPARATE writing booklet

- a)** The rate of flow (R litres per minute) of water through a filter is given by

$$R = 6t - \frac{t^2}{3}$$

- i)** At what times is no water flowing through the filter **2**
- ii)** Find the volume of the water that flows through the filter in the first 4 minutes **3**

- b)** In a chemical reaction the amount (M kilograms) of undissolved solid after t hours is given by

$$M = Ae^{-kt}.$$

After 24 hours 15% of the initial mass of solid remains undissolved.

- i)** The value of k **2**
- ii)** What percentage of solid has dissolved after 5 hours (give your answer correct to one decimal place) **2**
- iii)** How long it takes for 50% of the solid to be dissolved. (give your answer correct to the nearest minute) **2**

End of paper ☺

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Student Number



HSC Mathematics

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 3. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 A ☒ B ☒ C ☐ D ☐

Start Here



1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐

HSC Mathematics Task 3 2016 Marking Scheme

Question 4

(a) Differentiate with respect to x and fully simplify

i) $y = e^x \log_e x$

2

1 mark – correctly identifies need to use product rule 1 mark – finds the correct derivative from product rule	
$y = e^x \log_e x$ $u = e^x \quad v = \log_e x$ $u' = e^x \quad v' = \frac{1}{x}$ $\therefore \frac{dy}{dx} = uv' + vu'$ $\therefore \frac{dy}{dx} = \frac{e^x}{x} + e^x \log_e x$	<i>Generally very well done</i>

ii) $y = \cos(\log_e x)$

2

1 mark – gives $\sin(\log_e x)$ as the solution or a minor error 2 mark – finds the correct derivative from chain rule	
$y = \cos(\log_e x)$ $\frac{dy}{dx} = -\sin(\log_e x) \times \frac{1}{x} = -\frac{\sin(\log_e x)}{x}$	<i>Generally very well done</i>

(b) Evaluate using exact values $\int_1^{e^2} \frac{4}{x} dx$

3

1 mark – finds correct primitive 1 mark – substitutes limits correctly 1 mark – finds correct exact value	
$\int_1^{e^2} \frac{4}{x} dx = 4[\ln x]_1^{e^2}$ $= 4[\ln e^2 - \ln 1]$ $= 4[2 - 0]$ $= 8$	<i>Some students got mixed up with the constant – but generally well done</i> <i>May did not simplify to the whole number</i>

(c) Solve for x

3

$$e^{2x} - 2e^x = 0$$

1 mark – factorises as a quadratic 1 mark – discounts $e^x = 0$ with a reason 1 mark – correct solution $x = \log_e 2$	
$e^{2x} - 2e^x = 0$ $\therefore e^x(e^x - 2) = 0$ $\therefore e^x = 0$ no solution since $e^x > 0$ OR $\therefore e^x = 2$ $x = \log_e 2$	<i>Many students did not create a quadratic</i> <i>Many students did not say why the solution $e^x = 0$ and lost a mark</i>

(d) Let $y = \log_e(x - 1)$

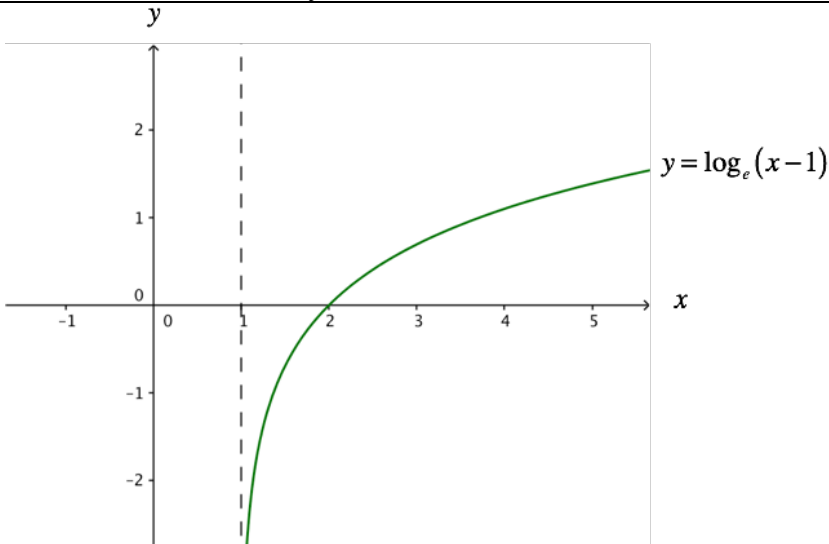
i) Find the x – intercept.

2

1 mark – correct x intercept 1 mark – sufficient working	
$y = \log_e(x - 1)$ $\therefore 0 = \log_e(x - 1)$ $\therefore x = 2$ x intercept is $(2, 0)$	<i>Genearily very well done</i>

ii) Sketch the graph of $y = \log_e(x - 1)$

2

1 mark – x intercept and asymptote clearly and correctly indicated 1 mark – correct shape of graph lose 1 mark if not evenly marked scale	<i>Many did not label the asymptote or have the correct one</i> <i>Many had no even scale or their graph did not work on the scale they had given</i>
	

- iii) Find the exact value of the area bounded by $y = \log_e(x-1)$, x -axis and the line $x = 5$.

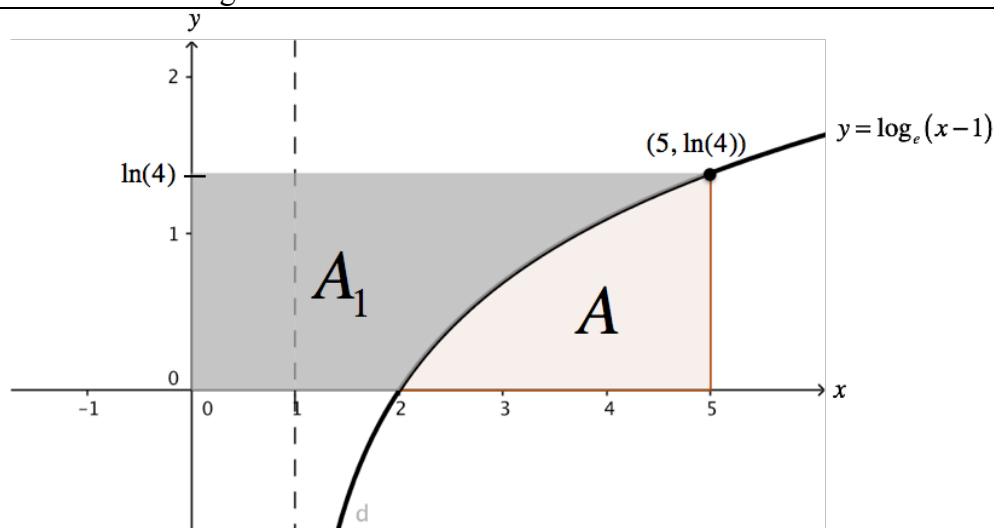
4

1 mark – breaking into rectangle with two separate areas

1 mark - making x the subject if $y = \log_e(x-1)$
 $\therefore x = e^y + 1$

1 mark – finding correct primitive and expression for the area

1 mark - finding exact value of area



*Not very well done
 Many tried to integrate log
 Some found A_1 but did not
 subtract from rectangle*

if $y = \log_e(x-1)$

$\therefore x = e^y + 1$

$$\int_2^5 \log_e(x-1) dx = A_{\text{rectangle}} - A_1$$

$$= (\ln(4) \times 5) - \int_0^{\ln 4} e^y + 1 dy$$

$$= 5 \ln(4) - [e^y + y]_0^{\ln 4}$$

$$= 5 \ln 4 - [(4 + \ln 4) - (1 - 0)]$$

$$= (4 \ln 4 - 3) \text{ units}^2$$

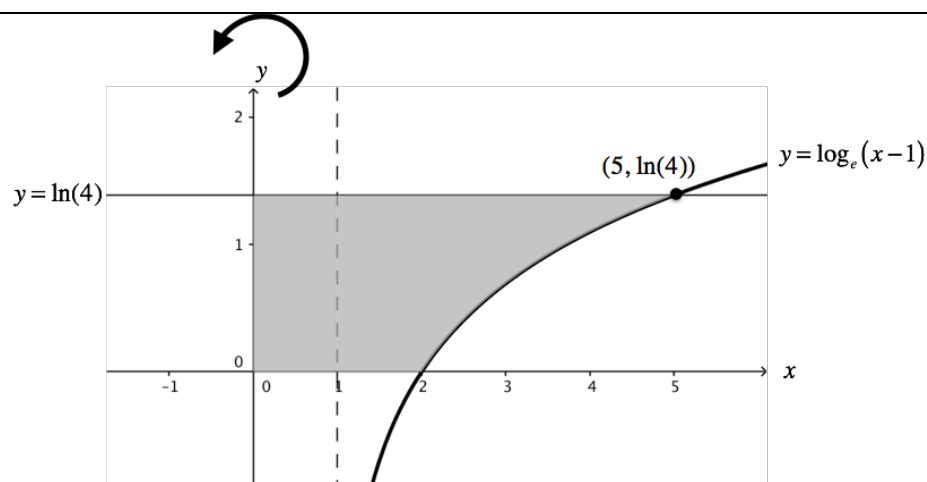
- iv) Find the exact volume of the solid generated when the area between $y = \log_e(x-1)$, the line $y = \log_e 4$, the x -axis and the y -axis is rotated about the y -axis.

3

1 mark: $V = \pi \int_0^{\ln 4} (e^y + 1)^2 dy$

1 mark: correct primitive for their integral (unless trivial)

1 mark: correct value for their integral



Many did not square or change the subject

Many did not square correctly

Generally poorly done

$$\begin{aligned}
 V &= \pi \int_0^{\ln 4} x^2 dy \\
 &= \pi \int_0^{\ln 4} (e^y + 1)^2 dy \\
 &= \pi \int_0^{\ln 4} (e^{2y} + 2e^y + 1) dy \\
 &= \pi \left[\frac{1}{2} e^{2y} + 2e^y + y \right]_0^{\ln 4} \\
 &= \pi \left[\left(\frac{e^{2\ln 4}}{2} + 2e^{\ln 4} + \ln 4 \right) - \left(\frac{1}{2} + 2 + 0 \right) \right] \\
 &= \pi \left[\left(\frac{16}{2} + 2 \times 4 + \ln 4 \right) - \left(\frac{5}{2} \right) \right] \\
 &= \pi \left(\frac{27}{2} + \ln 4 \right)
 \end{aligned}$$

Question 5

- a) Two standard six sided dice are tossed. Given that a 4 is showing on the upmost face of one of the dice, what is the probability that the sum of the upmost faces on both dice exceeds 8? 2

1 mark – lists the sample space in any format 1 mark – finds the correct probability																																																									
The sample space for a 4 showing on one upmost face Dice 1 <table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Dice 2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td></td><td></td><td></td><td>5</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td><td>6</td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td><td>7</td><td></td><td></td></tr><tr><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>5</td><td></td><td></td><td></td><td>9</td><td></td><td></td></tr><tr><td>6</td><td></td><td></td><td></td><td>10</td><td></td><td></td></tr></table> $\therefore P(\text{exceeds } 8) = \frac{4}{11}$		1	2	3	4	5	6	Dice 2							1				5			2				6			3				7			4	5	6	7	8	9	10	5				9			6				10			
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- b) In a basketball shootout, a player gets one attempt at scoring a 3 point shot and three attempts at scoring from a one point shot. From past statistics a players has a probability of 0.2 of scoring a three point shot and a probability of 0.75 of scoring a one-point shot.

- i) What is the probability that a player scores 0 points 1

1 mark: correct probability with working	
$P(0 \text{ points}) = 0.8 \times (0.25)^3 = 0.0125$	

- ii) What is the probability that a player scores all three one point shots but misses the three point shot? 1

1 mark: correct probability with working	
$P(\text{score three 1 points but miss 3 point}) = 0.8 \times (0.75)^3 = 0.3375$	

- iii) What is the probability that, a player scores at least 2 points? 3

1 mark: recognises complimentary event or lists possible outcomes 1 mark: correctly calculates the probability for 1 point 1 mark: obtains correct probability	
$ \begin{aligned} P(\text{at least 2 points}) &= 1 - (P(0 \text{ points}) + P(1 \text{ point})) \\ &= 1 - (0.0125 + (0.8 \times 0.75 \times (0.25)^2 \times 3)) \\ &= 0.959 \end{aligned} $	

Question 6

- a) The rate of flow (R litres per minute) of water through a filter is given by

$$R = 6t - \frac{t^2}{3}$$

- i) At what times is no water flowing through the filter

2

1 mark: equates derivative to 0 1 mark: correctly finds both times	
we need $R = \frac{dV}{dt} = 6t - \frac{t^2}{3} = 0$ $\therefore 18t - t^2 = 0$ $\therefore t(18 - t) = 0$ $\therefore t = 0$ minutes or $t = 18$ minutes	

- ii) Find the volume of the water that flows through the filter in the first 4 minutes

3

1 mark: establishes correct integral 1 mark: finds correct primitive function 1 mark: finds correct value and gives units	
Since $\frac{dV}{dt} > 0$ for $0 \leq t \leq 4$ $V(\text{water first 4 min}) = \int_0^4 \left(6t - \frac{t^3}{3} \right) dt$ $= \left[3t^2 - \frac{t^4}{12} \right]_0^4$ $= \left[3(4)^2 - \frac{4^4}{12} \right] - [0]$ $= \frac{368}{3} \text{ litres}$	

- b)** In a chemical reaction the amount (M kilograms) of undissolved solid after t hours is given by

$$M = Ae^{-kt}.$$

After 24 hours 15% of the initial mass of solid remains undissolved.

- i)** The value of k

2

1 mark: substituting into expression to obtain $0.15A = Ae^{-24k}$ 1 mark: finds correct value of k (any number of decimal places) with working	
When $t = 24$, $M = 0.15A$ $\therefore 0.15A = Ae^{-24k}$ $\therefore 0.15 = e^{-24k}$ $\therefore \log_e(0.15) = -24k$ $\therefore k = -\frac{\log_e(0.15)}{24}$ $\therefore k = 0.07904666604$	

- ii)** What percentage of solid has dissolved after 5 hours (give your answer correct to one decimal place)

2

1 mark: finding $\therefore M = 0.6735228677A$ with working 1 mark: correctly concluding the amount of the original solid that has dissolved	
When $t = 5$, $k = -\frac{\log_e(0.15)}{24}$, $M = ?$ $\therefore M = Ae^{-5 \times \left(\frac{\log_e(0.15)}{24}\right)}$ $\therefore M = 0.6735228677A$ (this is the amount remaining) $\therefore 32.65\%$ of the original solid has dissolved	

- iii)** How long it takes for 50% of the solid to be dissolved. (give your answer correct to the nearest minute)

2

1 mark: finding $\therefore \log_e(0.5) = -\left(\frac{\log_e(0.15)}{24}\right)t$ 1 mark: correctly identifying the correct time (to the nearest minute)	
When $t = ?$, $k = -\frac{\log_e(0.15)}{24}$, $M = 0.5A$ $\therefore 0.5A = Ae^{-\left(\frac{\log_e(0.15)}{24}\right)t}$ $\therefore \log_e(0.5) = -\left(\frac{\log_e(0.15)}{24}\right)t$ $\therefore t = 8$ hours 46 minutes	